

Mark schemes

Q1.

(a) (i) $\int \frac{dy}{y} = \int \sin t \, dt$

Attempt to separate and integrate

M1

$$\ln y = -\cos t + c$$

A1 for $\ln y$; A1 for $-\cos t$; condone missing c

A1,A1

$$y = ae^{-\cos t}$$

A present; or $y = e^{-\cos t + C}$

A1

4

(ii) $y = 50, t = \pi: 50 = Ae^{-\cos \pi} = Ae$

Substitute $y = 50, t = \pi$ to find constant

Can have $50 = e^{1+C}$ if substituted in above

$$e^C = \frac{50}{e}$$

M1

A1

$$y = 50e^{-1}e^{-\cos t}$$

AG (convincingly obtained)

A1

3

Alternative:

Must have a constant in answer to (a)(i)

$$y = Ae^{-\cos t} \text{ or } y = e^{-\cos t + c} \text{ or } \ln y = -\cos t + c$$

$$50 = Ae^{-\cos \pi} \quad 50 = e^{-\cos \pi + c} \quad \ln 50 = -\cos \pi + c$$

(M1)

$$50 = Ae \quad 50 = e^{1+c} \quad \ln y = -\cos t + \ln 50 - 1$$

(A1)

$$y = 50e^{-1-\cos t} \quad y = e^{-\cos t} \quad \frac{50}{e} \ln\left(\frac{y}{50}\right) = -1 - \cos t$$

(A1)

Alternative:

Substitute $y = 50$, $t = \pi$ into $\ln y = -\cos t + c$ M1

$\ln y = -\cos t + \ln 50 - 1$ A1

$$\ln \frac{y}{50} = -1 - \cos t \quad (AG) \quad A1$$

(b) (i) $t = 6: y = 50e^{-1}e^{-\cos 6} = 7.0417... \approx 7\text{cm}$

Degrees 6.8 SC1

7 or 7.0 for A1

M1A1

2

(ii) $t = \pi \Rightarrow (\sin t = 0 \Rightarrow) \frac{dy}{dt} = 0$

Condone x for t

B1

$$\frac{d^2 y}{dt^2} = y \cos t + \frac{dy}{dt} \sin t$$

For attempt at product rule including $\frac{dy}{dt}$

term; must have $\frac{d^2 y}{dt^2} =$

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$$t = \pi$$

A1

$$\frac{d^2 y}{dt^2} = y \cos \pi + \frac{dy}{dt} \sin \pi$$

$$= -50 \Rightarrow \max$$

Accept $= -y$, with explanation that y is never negative

A1

4

Alternative:

$$y = 50e^{-(1+\cos t)} = \frac{50}{e} e^{-\cos t}$$

$$\frac{dy}{dt} = \frac{50}{e} e^{-\cos t} \times \sin t = 0 \text{ at } t = \pi$$

(B1)

$$\frac{d^2y}{dt^2} = \frac{50}{e} e^{-\cos t} \times \cos t + \frac{50}{e} e^{-\cos t} \times \sin^2 t$$

Attempt at product rule

(M1)

Correct

(A1)

Substitute $t = \pi \rightarrow -50 \Rightarrow \text{max}$

(A1)

[13]

Q2.

(a) (i) $t^3 - 52t + 96$

one term correct

M1

another term correct

A1

all correct (no + c etc)

A1

3

(ii) $3t^2 - 52$

ft one term correct

M1

ft all "correct"

A1 ✓

2

(b) $\frac{dy}{dt} = 8 - 104 + 96$

substitute $t = 2$ into their $\frac{dy}{dt}$

M1

$= 0 \Rightarrow$ stationary value
 CSO; shown = 0 + statement

A1

Substitute $t = 2$ into $\frac{d^2y}{dt^2}$ ($= -40$)
 any appropriate test, e.g. $y'(1)$ and $y'(3)$

M1

$\frac{d^2y}{dt^2} < 0 \Rightarrow$
 max value
 all values (if stated) must be correct

A1

4

(c) Substitute $t = 1$ into their $\frac{dy}{dt}$
 must be their $\frac{dy}{dt}$ NOT $\frac{d^2y}{dt^2}$

M1

Rate of change = 45 (cm s^{-1})
 ft their $y'(1)$

A1 ✓

2

(d) Substitute $t = 3$ into their $\frac{dy}{dt}$
 interpreting their value of $\frac{dy}{dt}$

M1

$$(27 - 156 + 96 = 33 < 0)$$

$\Rightarrow$ decreasing when $t = 3$

allow increasing if their $\frac{dy}{dt} > 0$

E1 ✓

2

[13]

Q3.

(a) (i) $y = (x^2 - 3)e^x$

$$\frac{dy}{dx} = (x^2 - 3)e^x + 2xe^x$$

product rule

M1
A1

2

(ii) $\frac{d^2y}{dx^2} = (x^2 - 3)e^x + 2xe^x + 2xe^x + 2e^x$

product rule from their $\frac{dy}{dx}$

M1
A1

2

(b) (i) $\frac{dy}{dx} = 0$

$$\Rightarrow e^x(x^2 + 2x - 3) = 0$$

$$e^x f(x) = 0 \text{ from } \frac{dy}{dx} = 0$$

M1

$$e^x(x + 3)(x - 1) = 0$$

attempt at factorising or use of formula

m1

$$\therefore x = -3, 1$$

first correct solution

A1

second correct solution, and no others
SC No working shown:

$$x = -3 \quad B2, \quad x = 1 \quad B2$$

A1

4

(ii) $x = -3 \quad y'' = -4e^x \max \quad (-0.2)$
Condone slip

M1

$$x = 1 \quad y'' = 4e^x \min \quad (10.9)$$

A1

2

[10]

Q4.

(a) (i) $t = 0: x = 3$

B1

1

(ii) $t = 14: x = 15 - 12e^{-1}$

or $15 - 12e^{\frac{-14}{14}}$

$= 10.6$

CAO

M1

A1

2

(b) (i) $-5 = -12e^{-\frac{t}{14}}$

substitute $x = 10$; rearrange to form $p = qe^{-\frac{t}{14}}$

M1

$$\ln\left(\frac{5}{12}\right) = -\frac{t}{14} \quad (\text{OE})$$

take lns correctly

m1

$$t = 14 \ln\left(\frac{12}{5}\right)$$

must come from correct working

A1

3

(ii) $t = 12.256... \approx 12$ days

ft on a, b if $a > b$; accept $t = 12$ NMS

Accept 12 from incorrect working in (b)(i)

Accept 13 if 12.2 or 12.3 seen

B1F

1

(c) (i) $\frac{dx}{dt} = -\frac{1}{14}x - 12e^{-\frac{t}{14}}$

differentiate; allow sign error

condone $\frac{dy}{dx}$ used consistently

M1

$$= -\frac{1}{14}(x - 15)$$

Or $\frac{1}{14}\left(12e^{-\frac{t}{14}}\right)$ and $12e^{-\frac{t}{14}} = 15 - x$ seen

m1

$$= \frac{1}{14}(15 - x)$$

AG – be convinced CSO

A1

3

Alt: $t = -14 \ln\left(\frac{15 - x}{12}\right)$

attempt to solve given equation for t

(M1)

$$\frac{dt}{dx} = \frac{-14\left(-\frac{1}{12}\right)}{\left(\frac{15 - x}{12}\right)}$$

differentiate wrt x , with $\frac{1}{12}$ seen; OE

(m1)

$$\frac{dt}{dx} = \frac{14}{15-x} \Rightarrow \frac{dx}{dt} = \frac{1}{14}(15-x)$$

AG – be convinced

(A1)

(3)

Alt: (backwards)

$$\int \frac{dx}{15-x} = \int \frac{dt}{14} = \pm 14 \ln(15-x) = t + c$$

(M1)

Use (0,3): $-14 \ln(15-x) + 14 \ln 12 = t$

(m1)

Solve for x : $x = 15 - 12e^{-\frac{t}{14}}$

All steps shown

(A1)

(3)

(ii) rate of growth = 0.5 (cm per day)

Accept $\frac{7}{14}$

B1

1

[11]

Q5.

(a) (i) $\frac{dV}{dt} = 2t^5 - 8t^3 + 6t$

One term correct unsimplified

M1

Further term correct unsimplified

A1

All correct unsimplified (no + c etc)

A1

3

(ii) $\frac{d^2V}{dt^2} = 10t^4 - 24t^2 + 6$
One term FT correct unsimplified

M1

CSO. All correct simplified

A1

2

(b) Substitute $t = 2$ into their $\frac{dV}{dt}$

M1

$(= 64 - 64 + 12) = 12$

CSO. Rate of change of volume is $12 \text{ m}^3 \text{ s}^{-1}$

A1

2

(c) (i) $t = 1 \Rightarrow \frac{dV}{dt} = 2 - 8 + 6$
Or putting their $\frac{dV}{dt} = 0$

M1

$= 0 \Rightarrow$ Stationary value

CSO. Shown to $= 0$ **AND** statement
(If solving equation must obtain $t = 1$)

A1

2

(ii) $t=1 \Rightarrow \frac{d^2V}{dt^2} = -8$

Sub $t = 1$ into their second derivative or equivalent full test.

M1

Maximum value

Ft if their test implies minimum

A1 ✓

2

[11]

Q6.

(a) (i) When $x = 4$, $\frac{dy}{dx} = 3(2) + \frac{16}{16} - 7 = 0$

AG Be convinced

B1

1

(ii) $\frac{16}{x^2} = 16x^{-2}$

Accept $k = -2$

B1

1

(iii) $\frac{d^2y}{dx^2} = 3 \times \frac{1}{2} x^{-\frac{1}{2}} + 16 \times (-2)x^{-3} - 0$

A power decreased by 1

M1

$\frac{d^2y}{dx^2} = \frac{3}{2} x^{-\frac{1}{2}} ; -32x^{-3}$

candidate's negative integer k [-1 for >2 term(s)]

A1;
A1 ✓

3

(iv) When $x = 4$, $\frac{d^2y}{dx^2} = \frac{3}{4} - \frac{32}{64} = \frac{1}{4}$

Attempt to find $y''(4)$ reaching as far as two simplified terms

M1

Minimum since $y''(4) > 0$

candidate's sign of $y''(4)$

E1 ✓

2

[Alternative: Finds the sign of $y'(x)$ either side of the point where $x = 4$, need evidence rather than just a statement: (M1)

Correct ft conclusion with valid reason E1 ✓]

[In both, condone absent statement $y'(4) = 0$]

(b) (i) At $P(1,8)$, $\frac{dy}{dx} = 3(1)^{\frac{1}{2}} + \frac{16}{1^2} - 7 = 12$
AG Be convinced

B1

1

(ii) Gradient of normal = $-\frac{1}{12}$
Use of or stating $m \times m' = -1$

M1

Equation of normal is $y - 8 = m[x - 1]$
Can be awarded even if $m = 12$

M1

$y - 8 = -\frac{1}{12}(x - 1) \Rightarrow 12y - 96 = -x + 1 \Rightarrow 12y + x = 97$
Any correct form of the equation

A1

3

$\int 3x^{\frac{1}{2}} + \frac{16}{x^2} - 7dx = \dots = 3 \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + 16 \frac{x^{-1}}{-1} - 7x + c$

(c) (i) *One power correct.*

M1

*A1 if 2 of 3 terms correct candidate's negative integer k
 Condone absence of "+ c"*

A2,1,0 ✓

3

(ii) $y = 2x^{\frac{3}{2}} - 16x^{-1} - 7x + c$ (*)
 y = candidate's answer to (c)(i) with tidied coefficients and with '+c'. ('y =' PI by next line)

B1 ✓

When $x = 1, y = 8 \Rightarrow 8 = 2 - 16 - 7 + c$
Substitute. (1,8) in attempt to find constant of integration

M1

$$y = 2x^{\frac{3}{2}} - 16x^{-1} - 7x + 29$$

Accept $c = 29$ after (), including $y =$, stated*

A1

3

[17]

Q7.

(a) $y = e^{2x} - 10e^x + 12x$

(i) $\frac{dy}{dx} = 2e^{2x} - 10e^x + 12$
 $2e^{2x}$

B1

remaining terms correct, no extras

B1

2

(ii) $\frac{d^2y}{dx^2} = 4e^{2x} - 10e^x$
ft 1 slip

B1F

1

(b) (i) $2e^{2x} - 10e^x + 12 = 0$
 $e^{2x} - 5e^x + 6 = 0$

AG (be convinced)

B1

1

(ii) $z^2 - 5z + 6 = 0$

use of $z = e^x$ oe

M1

$$z = 2, 3$$

$$z = 2, e^x = 2$$

$$x = \ln 2$$

finding $e^x =$ their 2,3

M1

$$z = 3, e^x = 3$$

$$x = \ln 3$$

all correct AG

A1

SC: verification

$$\ln 2 \quad (B1)$$

$$\ln 3 \quad (B1)$$

3

(iii) $x = \ln 2$:

$$y = e^{2\ln 2} - 10e^{\ln 2} + 12\ln 2$$

either substitution of their $x = \ln 2$ ($e^x = 2$)

or their $x = \ln 3$ ($e^x = 3$)

M1

or

$$2^2 - 10 \times 2 + 12 \ln 2 = 4 - 20 + 12 \ln 2 = -16 + 12 \ln 2$$

A1

$$x = \ln 3 :$$

$$y = e^{2\ln 3} - 10e^{\ln 3} + 12\ln 3 = 9 - 30 + 12 \ln 3 = -21 + 12 \ln 3$$

A1

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(iv) $x = \ln 2$: ©2025 Exam Papers Practice. All rights reserved.

$$\frac{d^2 y}{dx^2} = 4e^{2\ln 2} - 10e^{\ln 2}$$

use of; in either of their $e^x = 2, 3$ into their $\frac{d^2 y}{dx^2}$

M1

$$= 16 - 20 = -4$$

$\therefore$ maximum

CSO

A1

$x = \ln 3$:

$$\frac{d^2y}{dx^2} = 4e^{2\ln 3} - 10e^{\ln 3}$$

$$= 36 - 30 = 6$$

$\therefore$ minimum

CSO

A1

3

[13]

Q8.

(a) Sides $24 - 2x$, $9 - 2x$

Either correct

B1

$$V = x(24 - 2x)(9 - 2x)$$

3 sides involving x multiplied together

M1

$$= 4x^3 - 66x^2 + 216x$$

AG (be convinced)

A1

3

(b) (i) $\frac{dV}{dx} = 12x^2 - 132x + 216$

Power decreased by 1

M1

One term correct

A1

All correct (no +C etc)

A1

3

(ii) Putting their $\frac{dV}{dx} = 0$ (must see this first)
Or their $12x^2 - 132x + 216 = 0$
Or $12(x^2 - 11x + 18) = 0$ or statement

M1

$$\Rightarrow x^2 - 11x + 18 = 0$$

AG (be convinced)

A1

2

(iii) $(x - 2)(x - 9) = 0$

M1

Factors, comp sq or formulae used (1 slip)

$x = 2, x = 9$

A1

2

(iv) Reject $x = 9$, since $9 - 2x < 0$

$x = 2$ is only possible value

E1

1

(c) (i) $\frac{d^2V}{dx^2} = 24x - 132$

Differentiating their $\frac{dV}{dx}$ (eg $2x - 11$)

Correct

M1

A1

2

(ii) $x = 2$ only $\Rightarrow \frac{d^2V}{dx^2} = -84$ (or < 0)

Correct $\frac{d^2V}{dx^2}$ value OE full test.

B1

Maximum value

ft if their test implies minimum

E1 ✓

2

[15]

Q9.

(a) (i) $p(3) = 27 - 45 + 21 - 3$

Finding $p(3)$

M1

$p(3) = 0 \Rightarrow x - 3$ is a factor
Shown = 0 plus a statement
Or $(x - 3)(x^2 - 2x + 1)$

A1

2

(ii) B is point (3,0)
Must have coordinates

B1

1

(b) (i) $\frac{dy}{dx} = 3x^2 - 10x + 7$
One term correct

M1

All correct with NO + c etc

A1

2

(ii) $3x^2 - 10x + 7 = 0$
Putting their $\frac{dy}{dx} = 0$

M1

$\Rightarrow (x - 1)(3x - 7) = 0$
Attempt to use quad formula or factorise

m1

$\Rightarrow$ at M , $x = \frac{7}{3}$
CSO factors correct etc

A1

3

(c) $\frac{d^2y}{dx^2} = 6x - 10$
ft their $\frac{dy}{dx}$

B1ft

$$\text{sub } x = 1, \Rightarrow \frac{d^2 y}{dx^2} = -4$$

$$\text{ft their } \frac{d^2 y}{dx^2}$$

B1ft

2

(d) (i) $\frac{x^4}{4} - \frac{5x^3}{3} + \frac{7x^2}{2} - 3x$ (+c)
Increase one power by 1

M1

One term correct

A1

Two other terms correct

A1

All correct (condone missing + c)

A1

4

- (ii) Realisation that limits are 0 and 1
Condone wrong way round

B1

$$\left[\frac{1}{4} - \frac{5}{3} + \frac{7}{2} - 3 \right] - 0$$

Attempt to sub their limits into their (d)(i)

M1

$$= -\frac{11}{12}$$

CSO. *Must use $F(1) - F(0)$ correctly*

A1

$$\text{Area} = \frac{11}{12}$$

CSO. *Convincing argument*

E1

4

[18]

Q10.

(a) $(2 + h)^3 = 8 + ah + bh^2 + h^3$

M1

$$(2 + h)^3 = 8 + 12h + 6h^2 + h^3$$

A1 for each of a, b; PI

A1A1

$$f(2 + h) - f(2) = 13h + 6h^2 + h^3$$

Ft one coeff. wrong

m1A1F

5

(b) Divide by h and let $h \rightarrow 0$

NMS B1F

M1

$$f'(2) = p = 13$$

ft wrong value of p

A1F

2

[7]